ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 2 · PRACTICE D · MARKING GUIDELINE

MARKS: 80

Marking guideline for the Academic Central NC(V) Level 2 Electronics and Digital Concepts for Robotics paper 2 · practice d.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Circuits and breadboard layouts may look different and still be correct: check that each connection is made. Deduct 1 mark for each wrong connection and 2 marks for any short circuit (no less than 0 for a question).

QUESTION 1

1.1 Battery with correct polarity ✓✓; slide switch in series with the supply ✓✓; three LEDs in series ✓✓✓, all facing the same (correct) direction ✓✓; one 220 Ω resistor in series with the LEDs ✓✓; buzzer ✓✓ connected across the whole LED-and-resistor string (in parallel) ✓✓ with correct polarity ✓; return to battery − ✓✓; neat layout and labels ✓✓ (20)

1.2 Placement, 1 mark each ✓✓✓✓✓✓✓✓✓ (battery, S1, R1, R2, R3, R4, Q1, LED1, LED2) = 9. Polarity ✓✓✓: LED1; LED2; transistor pins C-B-E = 3. Connections ✓✓✓✓✓✓✓✓: S1 from +9 V to the input point; R1 from the input point to 0 V; R3 and LED1 from the input point to 0 V; R2 from the input point to the base; emitter to 0 V; R4 from +9 V to the collector; LED2 from the collector to 0 V; battery to the rails = 8. (20)

[40]

QUESTION 2

2.1 Placement ✓✓✓✓✓✓✓ (battery, probes, R1, Q1, Q2, buzzer, R2 and LED – any seven) = 7. Polarity ✓✓✓: transistor pins of Q1 and Q2; LED; buzzer = 3. Connections ✓✓✓✓✓✓✓✓✓✓: probe P1 to +9 V; probe P2 to R1; R1 to the base of Q1; emitter of Q1 to the base of Q2; emitter of Q2 to 0 V; both collectors joined; buzzer from +9 V to the collectors; R2 from +9 V to the LED; LED to the collectors; battery to the rails = 10. (20)

2.2 Correct circuit diagram: 9 V battery ✓✓ → push button S1 ✓✓ → R1 1 kΩ ✓✓ → junction ✓✓; C1 from the junction to the − line ✓✓ with + towards the junction ✓✓; R2 from the junction ✓✓ to the LED anode ✓✓, LED cathode to the − line ✓✓; correct symbols and labels ✓✓ (20)

[40]

TOTAL: 80

END OF MARKING GUIDELINE